

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is: This message is generated by a gateway server that was waiting for a response from the web server. If the web server takes too long to respond, the gateway server will send a timeout error.

The logs show that: the same IP address 203.0.113.0 keeps sending SYN packets to the website, which slows down operations, causing actual visitors to receive a timeout message or RST/SYN error. Eventually, so many SYN packets are sent that the entire server just stops letting visitors synchronize with the website.

This event could be: A SYN flood attack, which is a type of DoS attack.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol.

Explain the three steps of the handshake:

1. First, the visitor sends a SYN packet to the web server to try to synchronize with it.
2. Next, the web server creates a SYN, ACK packet, which stands for synchronize, acknowledgement. This means the web server acknowledges the request to synchronise with the server.
3. Lastly, the visitor's machine creates an ACK packet, which is the acknowledgement of the connection between the host's machine and the web server. This is the final step to complete a full TCP connection.

Explain what happens when a malicious actor sends a large number of SYN packets all at once: When a malicious actor sends a large number of SYN packets, it floods the web server and overwhelms it, causing it to slow down, making it impossible for actual visitors to use the website.

Explain what the logs indicate and how that affects the server: The logs indicate that at first the server receives a large amount of SYN requests; this causes other users to receive a timeout error message. Eventually, so many SYN packets are sent that other visitors can't even send packets.